DeepSeek Coder (Zhu et al., 2024), and Llama-3.1 405b (Llama Team, 2024). The first two models are only available by a public API, whilst the second two models are open-weight. For each run, we provide 1-2 basic seed ideas as examples (e.g. modifying the learning rate or batch size) and have it generate another 50 new ideas. We visualize an example progression of proposed ideas in Appendix C. Each run of around fifty ideas in *total* takes approximately 12 hours on 8× NVIDIA H100s[2]. We report the number of ideas that pass the automated novelty check, successfully complete experiments, and result in valid compilable manuscripts. Note that the automated novelty check and search are self-assessed by each model for its own ideas, making relative "novelty" comparisons challenging. Additionally, we provide the mean and max reviewer scores of the generated papers and the total cost of the run. Finally, we select and briefly analyze some of the generated papers, which are listed below. The full papers can be found in Appendix D, alongside the generated reviews and code.

In practice, we make one departure from the formal description of THE AI SCIENTIST, and generate ideas without waiting for paper evaluations to be appended to the archive in order to parallelize more effectively. This allowed us to pay the cost of the idea generation phase only once and iterate faster; furthermore, we did not observe any reduction in the quality of the papers generated as measured by the average review score with this modification.

Table 2 | **10 selected papers generated by THE AI SCIENTIST across 3 different templates, together with scores from our automated reviewer corresponding to the NeurIPS guidelines. The average accepted paper at NeurIPS has a score of around 6 from human evaluation.**

Type	Paper Title	Score
2D Diffusion	DualScale Diffusion: Adaptive Feature Balancing for Low-Dimensional Generative Models	**5**
2D Diffusion	Multi-scale Grid Noise Adaptation: Enhancing Diffusion Models For Low-dimensional Data	**4**
2D Diffusion	GAN-Enhanced Diffusion: Boosting Sample Quality and Diversity	**3**
2D Diffusion	DualDiff: Enhancing Mode Capture in Low-dimensional Diffusion Models via Dual-expert Denoising	**5**
NanoGPT	StyleFusion: Adaptive Multi-style Generation in Character-Level Language Models	**5**
NanoGPT	Adaptive Learning Rates for Transformers via Q-Learning	**3**
Grokking	Unlocking Grokking: A Comparative Study of Weight Initialization Strategies in Transformer Models	**5**
Grokking	Grokking Accelerated: Layer-wise Learning Rates for Transformer Generalization	**4**
Grokking	Grokking Through Compression: Unveiling Sudden Generalization via Minimal Description Length	**3**
Grokking	Accelerating Mathematical Insight: Boosting Grokking Through Strategic Data Augmentation	**5**

From manual inspection, we find that Claude Sonnet 3.5 consistently produces the highest quality papers, with GPT-4o coming in second. We provide a link to all papers, run files, and logs in our GitHub repository, and recommend viewing the uploaded Claude papers for a qualitative analysis. This observation is also validated by the scores obtained from the LLM reviewer (Figure 4). When dividing the number of generated papers by the total cost, we end up at a cost of around $10-15 per paper. Notably, GPT-4o struggles with writing LaTeX, which prevents it from completing many of its papers. For the open-weight models, DeepSeek Coder is significantly cheaper but often fails to correctly call the Aider tools. Llama-3.1 405b performed the worst overall but was the most convenient to work with, as we were frequently rate-limited by other providers. Both DeepSeek Coder and Llama-3.1 405b often had missing sections and results in their generated papers. In the following subsections, we will describe each template, its corresponding results, and specific papers.

6.1. Diffusion Modeling

General Description: This template studies improving the performance of diffusion generative models (Ho et al., 2020; Sohl-Dickstein et al., 2015) on low-dimensional datasets. Compared to image

[2]Note that the experiment templates are very small-scale and are not compute-intensive. They would likely take a similar amount of time on cheaper GPUs, as we do not achieve high utilization.

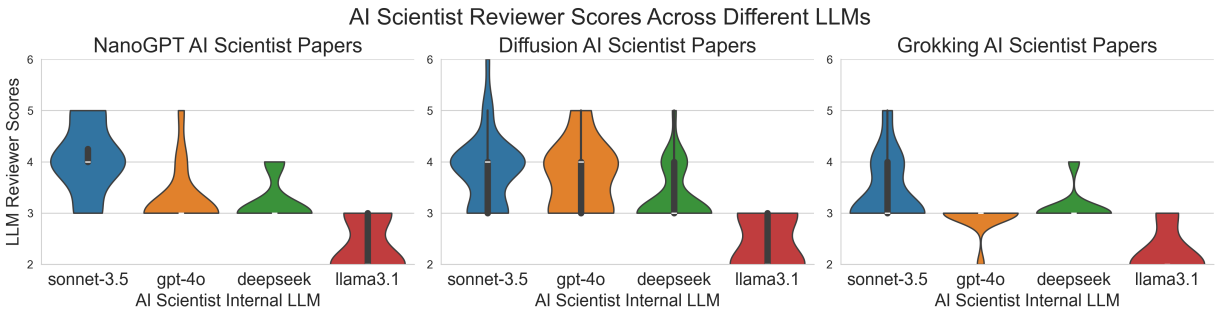


Figure 4 | Violin plots showing the distribution of scores generated by the THE AI SCIENTIST reviewer for AI-generated papers across three domains and four foundation models. Scores on the y-axis refer to NeurIPS ratings, which range from 2 (Strong Reject) to 6 (Weak Accept).

Table 3 | **Evaluation of automated AI Scientist paper generation for Diffusion Modeling.**

	Total Ideas	Novel Ideas	Experiments Passed	Completed Papers	Mean Score	Max Score	Total Cost
Sonnet 3.5	51	49	38	38	3.82	6.0	~$250
GPT-4o	51	41	17	16	3.70	5.0	~$300
DeepSeek Coder	51	42	32	31	3.32	5.0	~$10
Llama-3.1 405b	51	31	21	21	2.30	3.0	~$120

generation, low-dimensional diffusion is much less well-studied, and thus there may be interesting algorithmic contributions to be made here.

Code Template: We base this template on a modified version of the popular 'tanelp/tiny-diffusion' repository (Pärnamaa, 2023) with additional minor hyperparameter tuning added and exponential moving average on the weights. The diffusion models are DDPM (Ho et al., 2020) models trained to generate samples from four distributions including geometric shapes, the two moons dataset, and a 2D dinosaur. The denoiser network is parameterized as an MLP with sinusoidal embeddings for the diffusion timestep and input data. The plotting script visualizes generated samples and plots training loss by default. Estimated KL is provided as an additional metric for sample quality via non-parametric entropy estimation.

Highlighted Generated Paper 1: DualScale Diffusion: Adaptive Feature Balancing for Low-Dimensional Generative Models. We analyze this paper in-depth in Section 5. This paper proposes a dual-scale denoising approach that splits the traditional diffusion denoiser into a global and a local processing branch. The network input is upscaled before being fed into the local branch. The outputs of the branches are then combined using a learnable time-conditioned weighting. It achieves impressive quantitative and qualitative results. It further manages to plot the evolution of the weighting across time, which requires very significant deviation from the provided code.

Highlighted Generated Paper 2: Multi-scale Grid Noise Adaptation: Enhancing Diffusion Models For Low-dimensional Data. This paper proposes to dynamically scale the standard diffusion noise schedule with a learned multiplicative factor based on where a particular input is in 2D space. The multiplicative factor is set by two grids that cover the input space, one coarse 5x5 grid and one more fine-grained 20x20 grid. This creative approach allows the diffusion model to dramatically improve performance across the datasets.

Highlighted Generated Paper 3: GAN-Enhanced Diffusion: Boosting Sample Quality and Diversity. This paper, inspired by GANs, proposes adding a discriminator to the diffusion model to guide the generation. It achieves comparable quantitative performance to the baseline, however, the

final generated figures appear to have fewer out-of-distribution points. This is notable as the current version of THE AI SCIENTIST is unable to view them (a problem that can be remedied by using multi-modal models in the future).

Highlighted Generated Paper 4: DualDiff: Enhancing Mode Capture in Low-dimensional Diffusion Models via Dual-expert Denoising. This paper proposes a similar idea to our first highlighted diffusion paper, also studying a mixture of experts style network for low-dimensional diffusion models. However, this idea evolves differently, with the standard diffusion loss now being augmented with a loss that encourages diversity in the two experts. The paper impressively visualizes the impact of the diversity loss in distributing inputs across both experts and further color-codes which parts of the sample space each expert is specialized in. We were particularly impressed by THE AI SCIENTIST's ability to perform a radically different take on a similar idea.

6.2. Language Modeling

Table 4 | **Evaluation of automated AI Scientist paper generation for Language Modeling.**

	Total Ideas	Novel Ideas	Experiments Passed	Completed Papers	Mean Score	Max Score	Total Cost
Sonnet 3.5	52	50	20	20	4.05	5.0	~$250
GPT-4o	52	44	30	16	3.25	5.0	~$300
DeepSeek Coder	52	37	23	23	3.21	4.0	~$10
Llama-3.1 405b	52	41	21	21	2.31	3.0	~$120

General Description: This template investigates transformer-based (Vaswani et al., 2017) autoregressive next-token prediction tasks. Because this task is widely studied and optimized, it is difficult for THE AI SCIENTIST to find significant improvements. There are some common failure modes for this template that result in impressive-looking, but deceptive results. For example, a few of its ideas effectively cheat by subtly leaking information from future tokens, which results in lower perplexity.

Code Template: The code is modified from the popular NanoGPT repository (Karpathy, 2022). The provided script template trains a small transformer language model on the character-level Shakespeare dataset (Karpathy, 2015), the enwik8 dataset (Hutter, 2006), and the text8 dataset (Mahoney, 2011). It runs three seeds on the Shakespeare dataset, and one each on the remaining ones. The code saves the runtime, validation losses, and train losses. The plotting script visualizes training curves by default.

Highlighted Generated Paper 1: StyleFusion: Adaptive Multi-style Generation in Character-Level Language Models. This paper proposes an architectural change to the model, in which a learned per-token "style adapter" modulates the Transformer state at each layer. The method achieves strong results and deserves further investigation, though we suspect that one reason it may work is that it is simply adding more parameters, which may trivialize the result. Furthermore, it omits some important implementation details in the writing, such as how the style loss labels are derived (which appear to be randomly assigned on each update step).

Highlighted Generated Paper 2: Adaptive Learning Rates in Transformers via Q-Learning. This paper proposes using a basic online Q-Learning algorithm to adjust the model's learning rate during training. The state consists of the current learning rate and validation loss, the action applies a small perturbation to the learning rate, and the reward is the negative change in validation loss. While the idea is creative, it seems inappropriate to use simple Q-Learning in this highly non-stationary and partially-observed environment. Nonetheless, it happens to achieve effective results.